DAQ Core

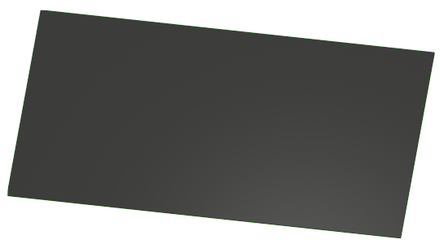
Variant: DRAFT

2026-11-03

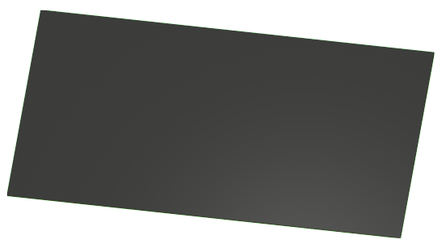
Rev + (Unreleased)

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2	Block Diagram	12	Command - High Power	22	Revision History	32	.....
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4	MCU	14	Interface - RS232 & RS485	24	.....	34	.....
5	MCU - Programming	15	Interface - CAN	25	.....	35	.....
6	MCU - Expander	16	Interface - Other	26	.....	36	.....
7	MCU - IOT	17	Sensor - Environment	27	.....	37	.....
8	Analog - Input 3V3	18	Sensor - IMU	28	.....	38	.....
9	Analog - Input 5V & 10V	19	Memory	29	.....	39	.....
10	Analog - Output	20	Power	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 03-Nov-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

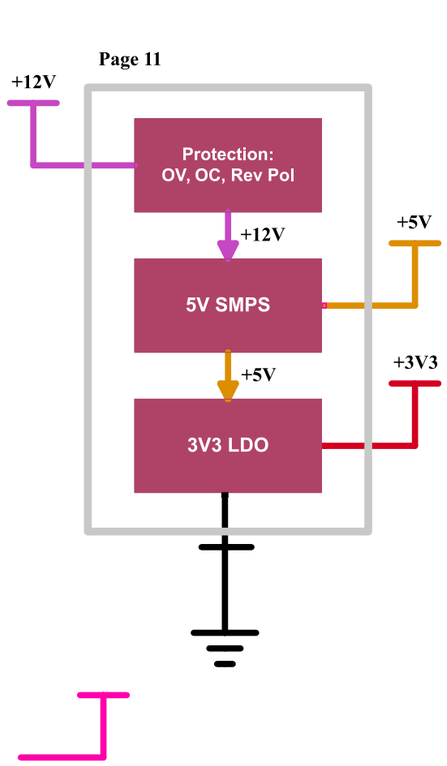
DESIGN NOTE: Example text for cautionary design notes.

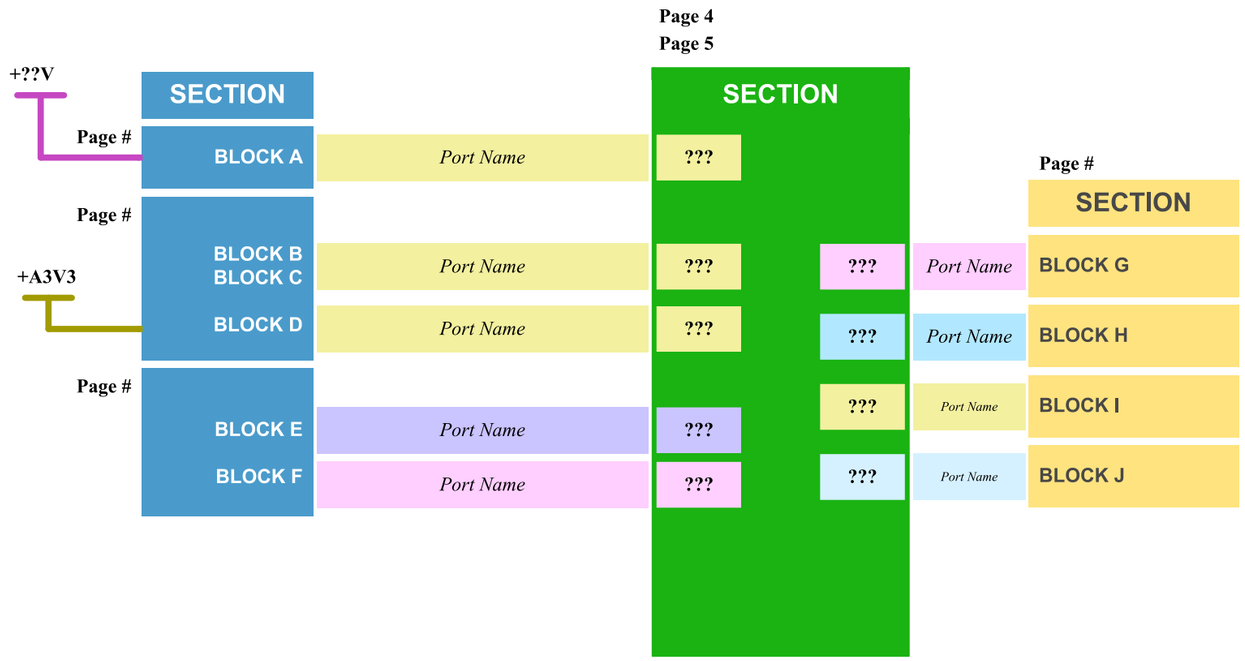
DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title:	File Name: Data-Acquisition-Board.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 22

[2] Block Diagram



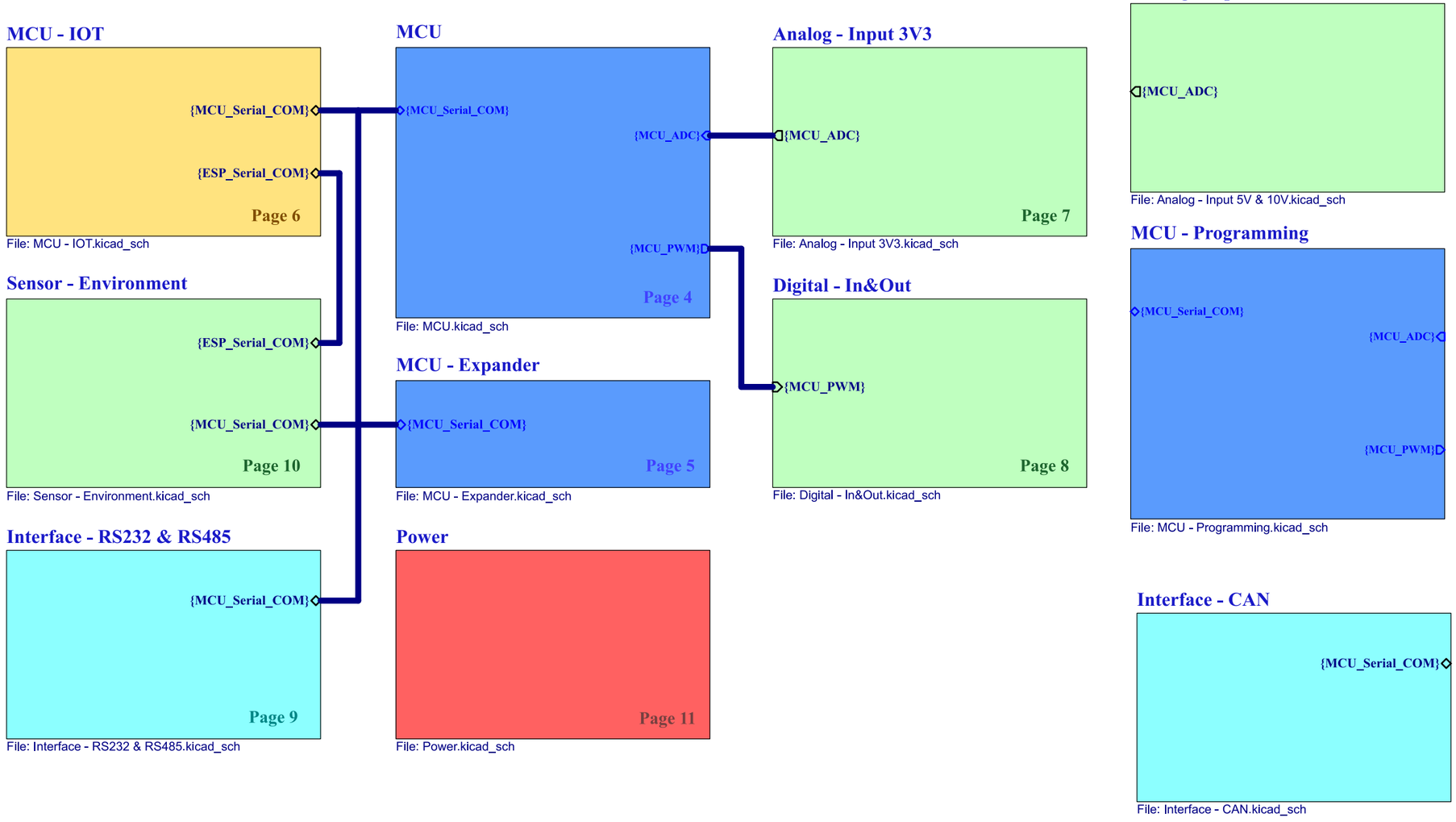


Target specifications:

Input voltage:	7V - 17V
Input power:	7400mW
Spec 3	??
Spec 4	??

Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **22**

[3] Project Architecture

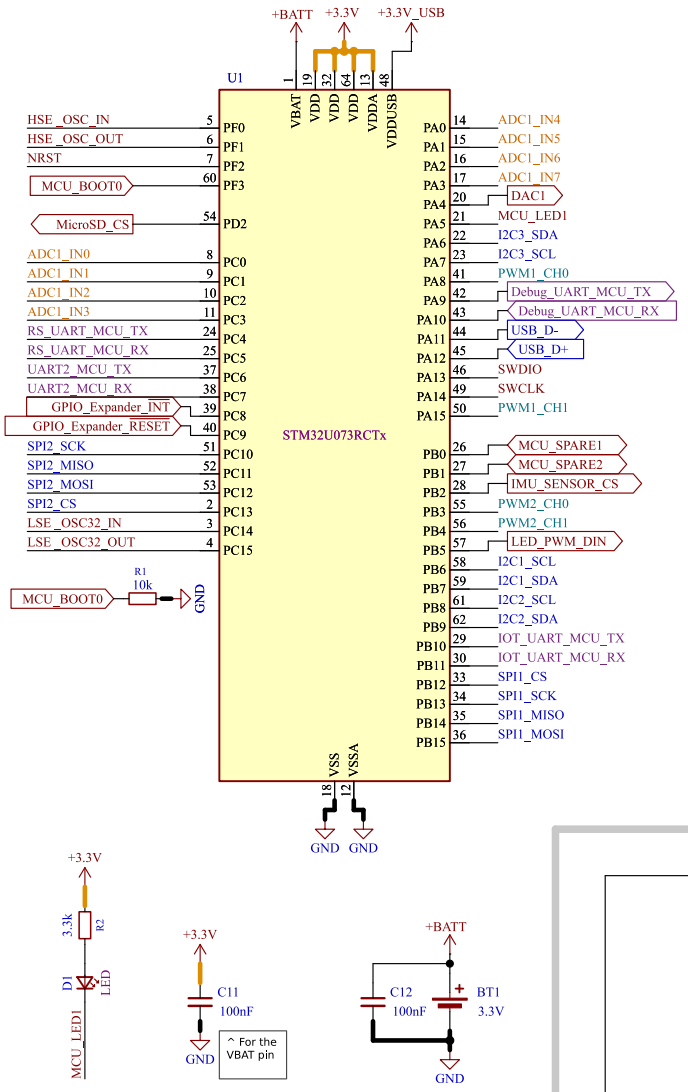


Description

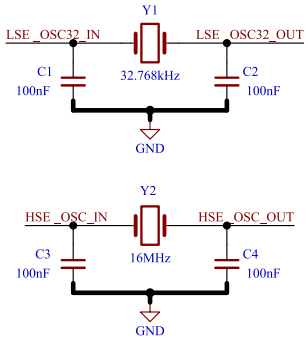
Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **22**

[4] MCU

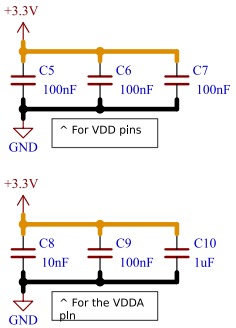
STM32U073RCTx



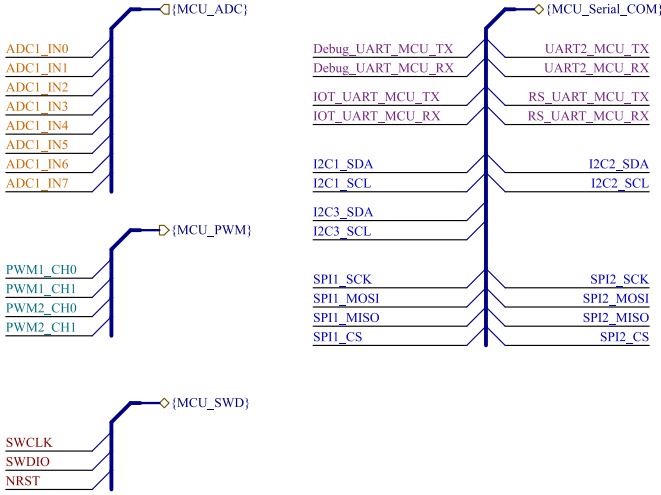
Quartz



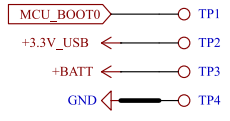
MCU Capacitor



BUS



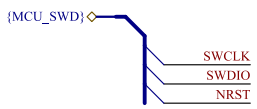
Test Points



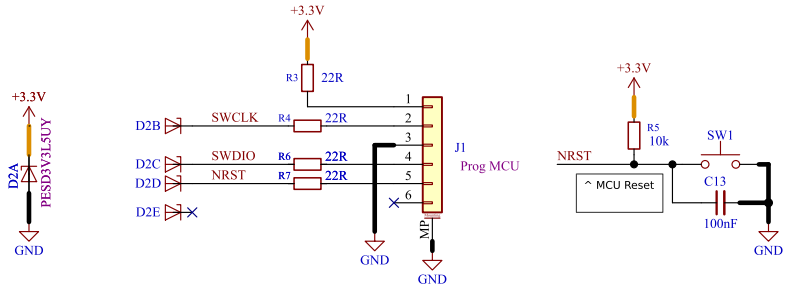
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: **A4**	Sheet: **4** of **22**

[5] MCU - Programming

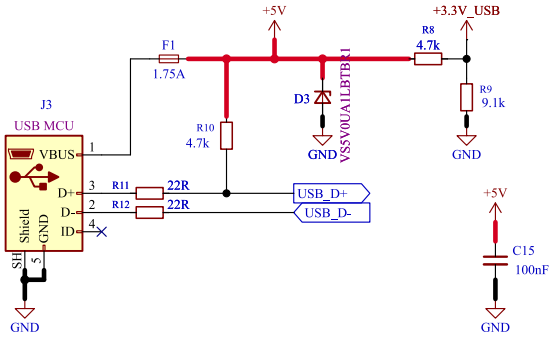
BUS



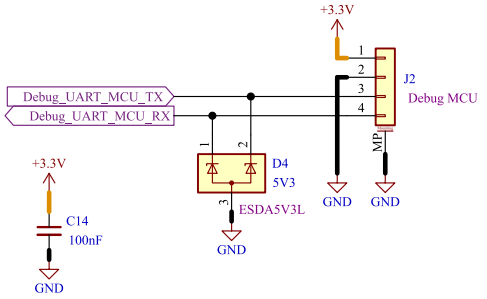
SWD Port



USB



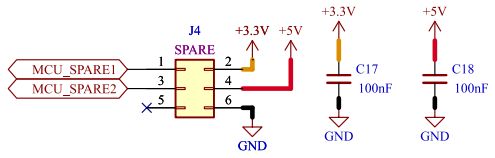
Debug



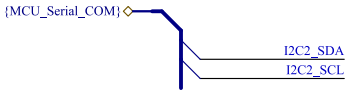
Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: MCU - Programming	File Name: MCU - Programming.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Programming/		Reviewer:	Size: **A4**	Sheet: **5** of **22**

[6] MCU - Expander

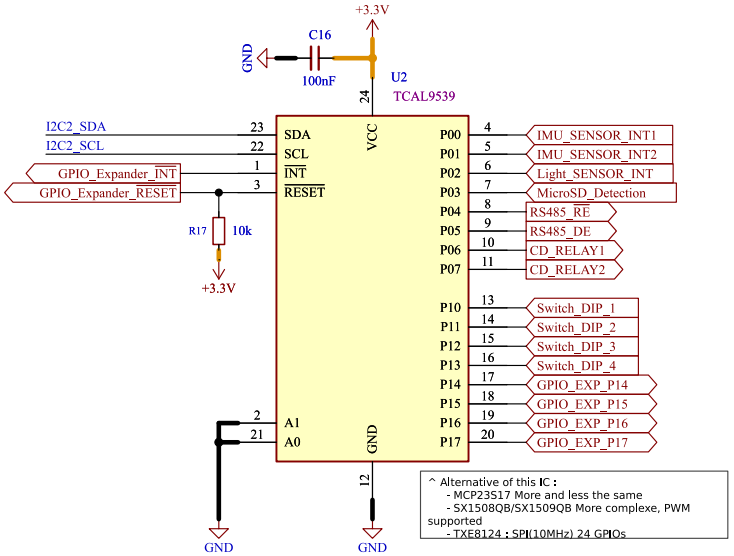
MCU SPARE CONNECTOR



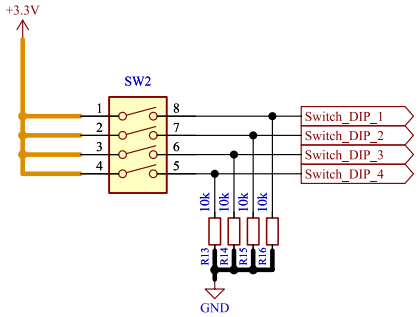
BUS



GPIO Expander



Switch DIP



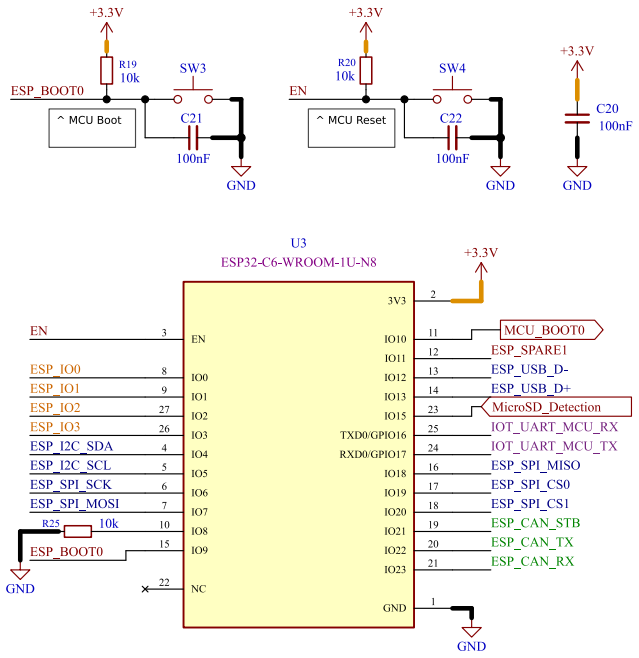
Test Points



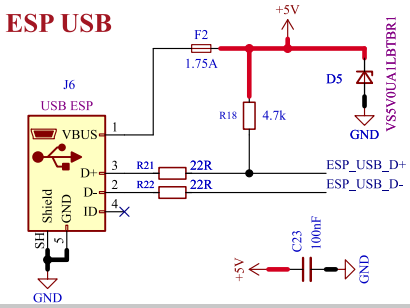
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU - Expander	File Name: MCU - Expander.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Expander/		Reviewer:	Size: A4	Sheet: 6 of 22

[7] MCU - IOT

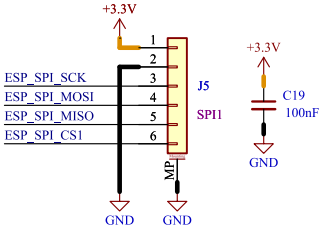
ESP32-C6-WROOM-1U



ESP USB



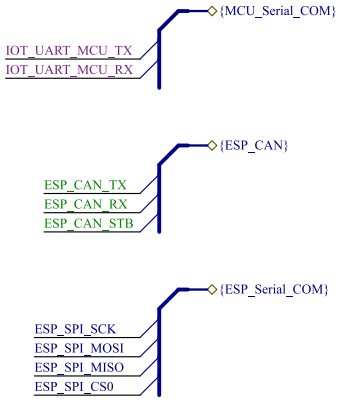
ESP SPI



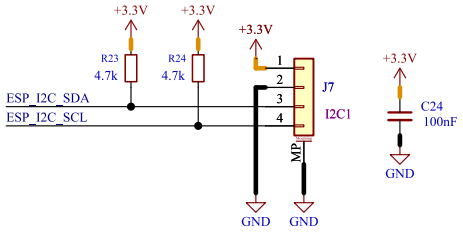
Test Points



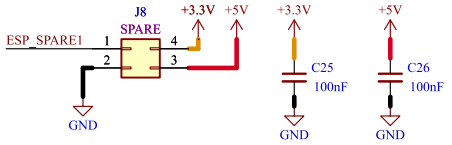
BUS



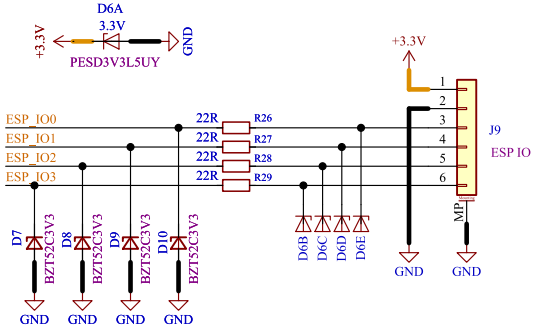
ESP I2C



ESP SPARE CONNECTOR



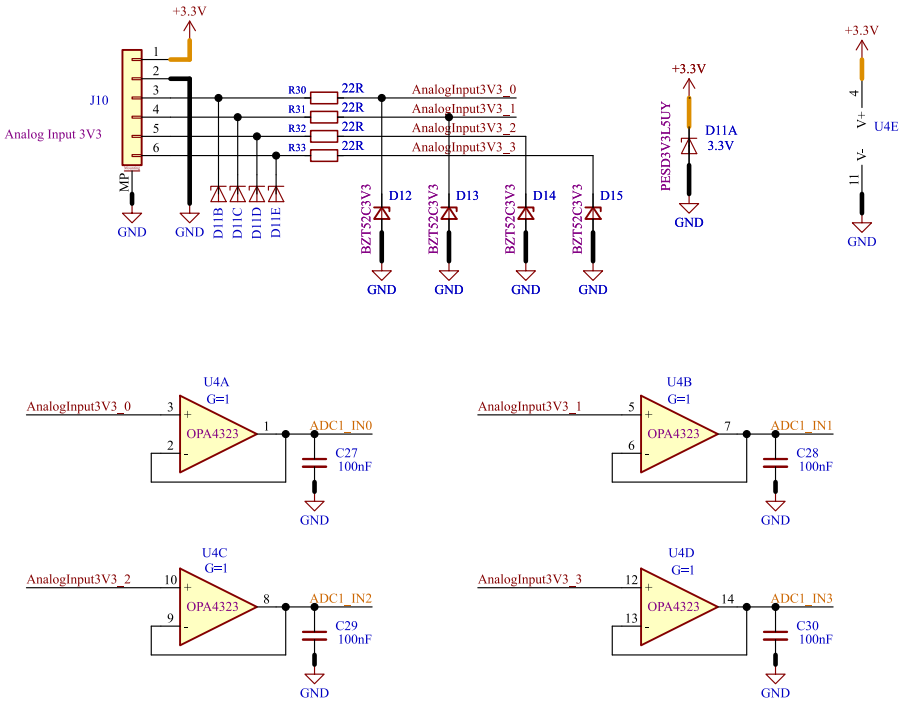
ESP IO



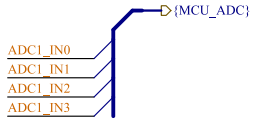
Comments: ESP32-C6-WROOM-1U Datasheet; TJA1042BT Datasheet	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU - IOT	File Name: MCU - IOT.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOT/		Reviewer:	Size: A4	Sheet: 7 of 22

[8] Analog - Input 3V3

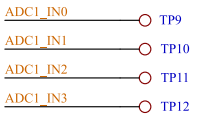
Analog Input ADC 3V3



BUS



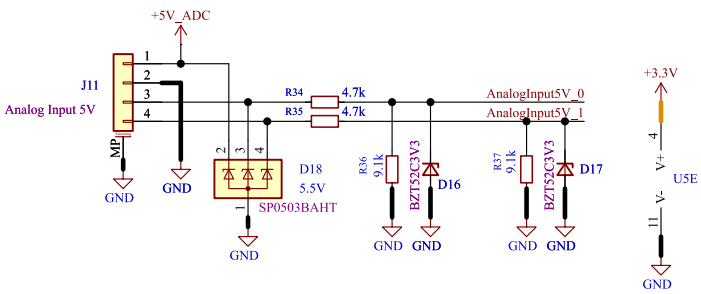
Test Points

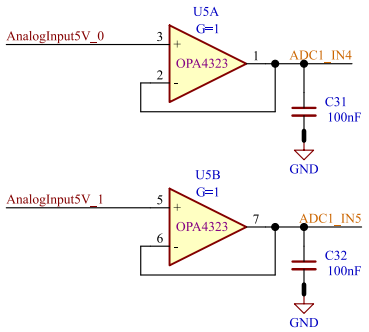


Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Analog - Input 3V3	File Name: Analog - Input 3V3.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Analog - Input 3V3/		Reviewer:	Size: A4	Sheet: 8 of 22

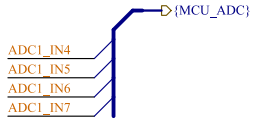
[9] Analog - Input 5V & 10V

Analog Input 5V

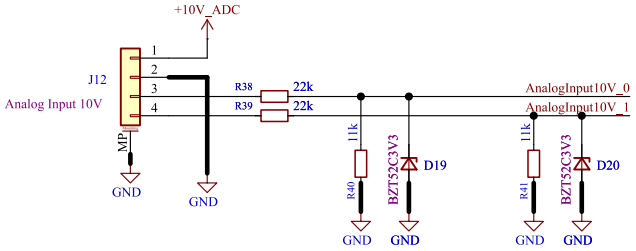


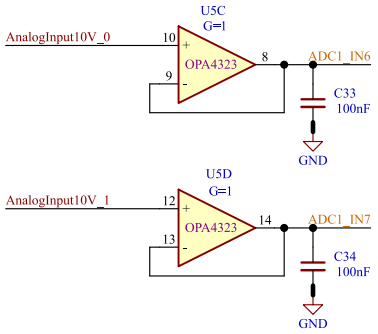


BUS

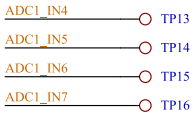


Analog Input 10V





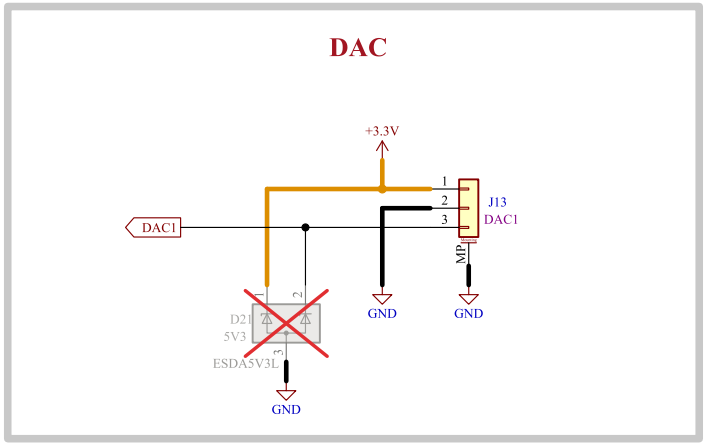
Test Points



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Analog - Input 5V & 10V	File Name: Analog - Input 5V & 10V.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Analog - Input 5V & 10V/		Reviewer:	Size: A4	Sheet: 9 of 22

[10] Analog - Output

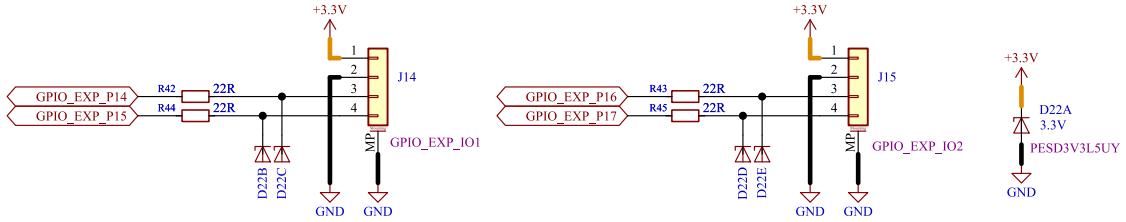
DAC



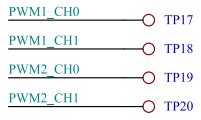
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Analog - Output	File Name: Analog - Output.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Analog - Output/		Reviewer:	Size: A4	Sheet: 10 of 22

[11] Digital - In&Out

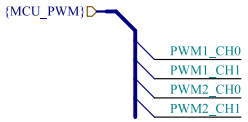
Digital I/O



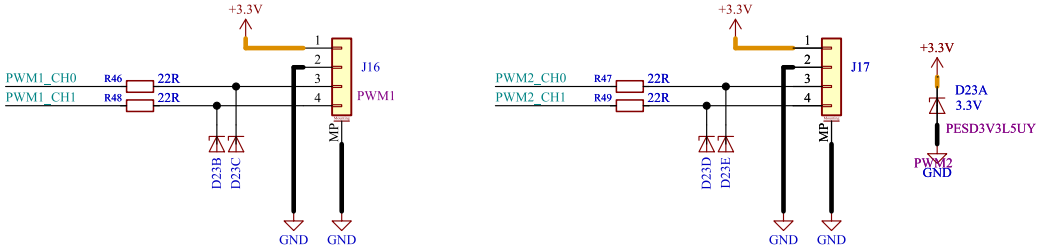
Test Points



BUS



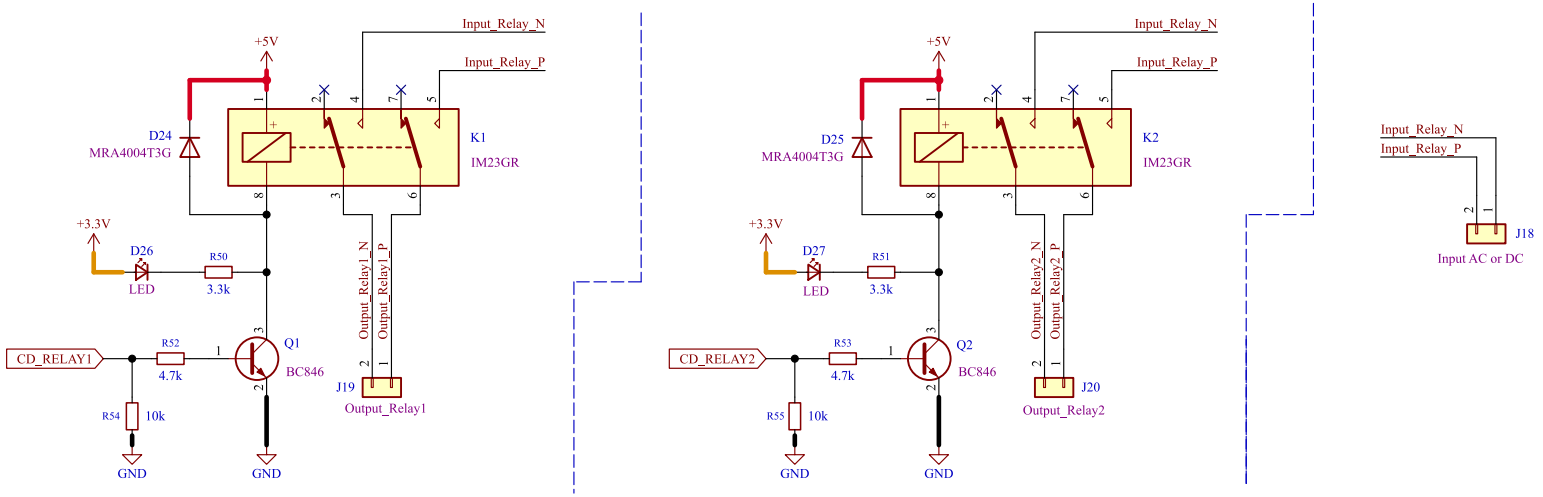
PWM



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Digital - In&Out	File Name: Digital - In&Out.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Digital - In&Out/		Reviewer:	Size: A4	Sheet: 11 of 22

[12] Command - High Power

Relay



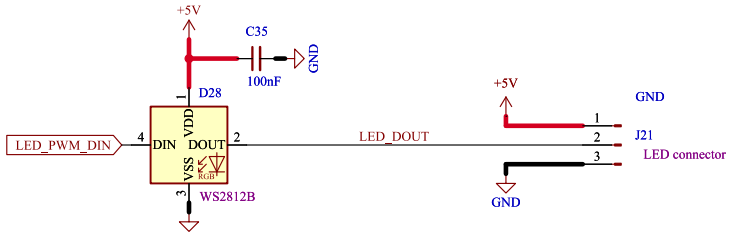
Test Points



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Command - High Power	File Name: Command - High Power.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Command - High Power/		Reviewer:	Size: A4	Sheet: 12 of 22

[13] Command - LED

LED strip



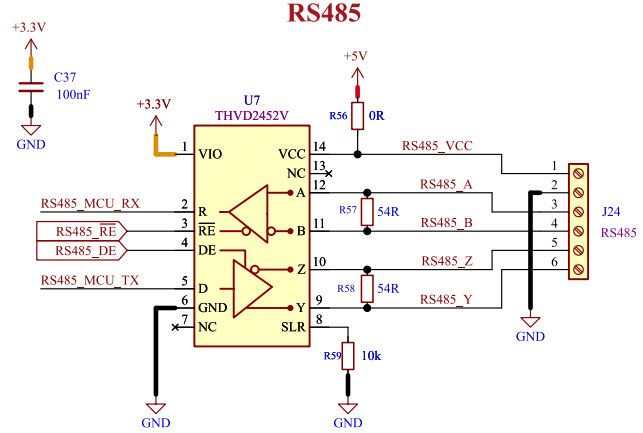
Test Points



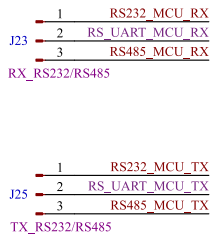
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Command - LED	File Name: Command - LED.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Command - LED/		Reviewer:	Size: A4	Sheet: 13 of 22

[14] Interface - RS232 & RS485

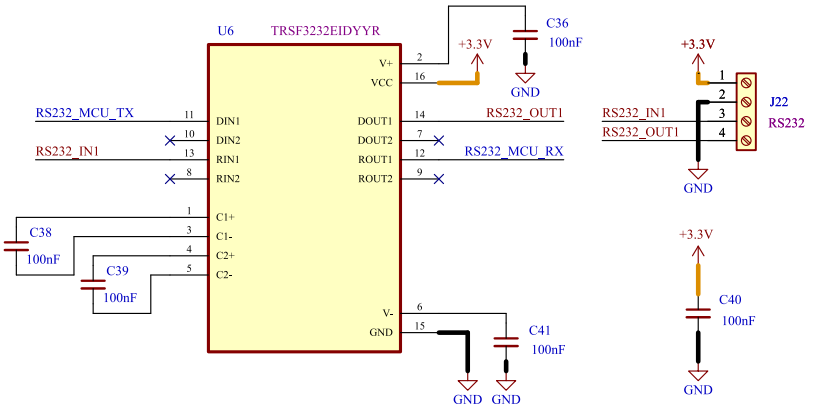
RS485



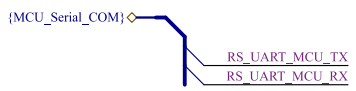
Selection RS232 or RS485



RS232



BUS

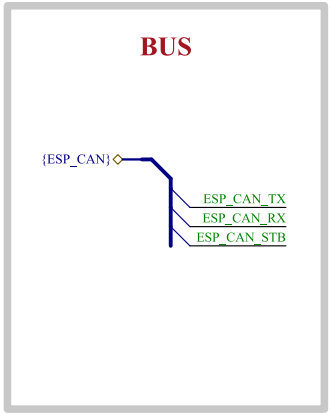


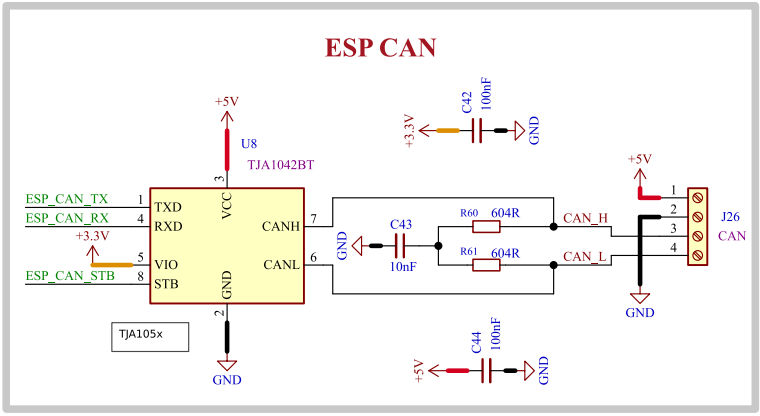
Test Points



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Interface - RS232 & RS485	File Name: Interface - RS232 & RS485.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - RS232 & RS485/		Reviewer:	Size: A4	Sheet: 14 of 22

[15] Interface - CAN

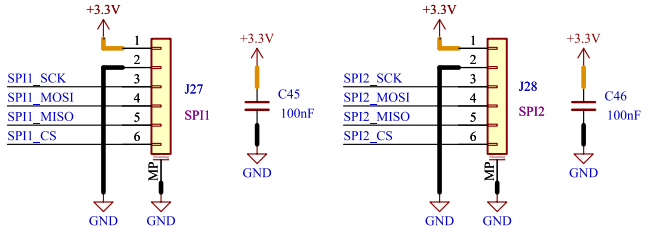




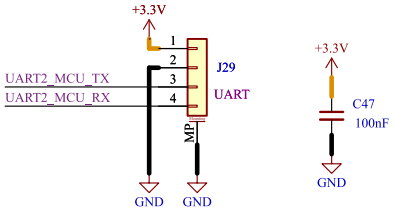
	Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
		Board Name: DAQ Core		Project Name: Data Acquisition Board	
	Sheet Title: Interface - CAN	File Name: Interface - CAN.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Interface - CAN/		Reviewer:	Size: A4	Sheet: 15 of 22

[16] Interface - Other

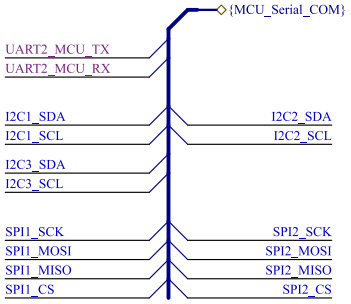
SPI



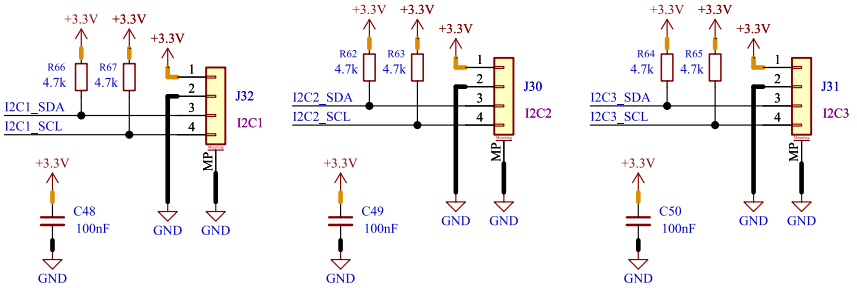
UART



BUS



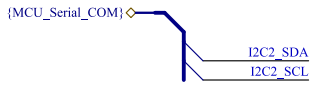
I2C



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Interface - Other	File Name: Interface - Other.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Other/		Reviewer:	Size: A4	Sheet: 16 of 22

[17] Sensor - Environment

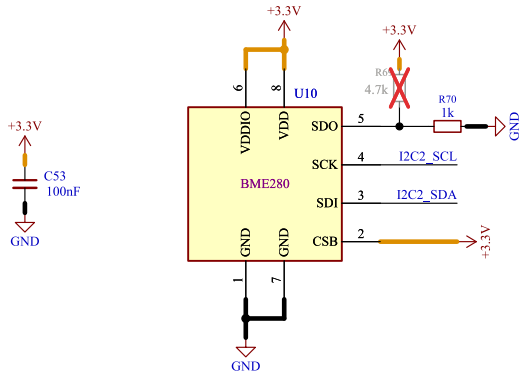
BUS



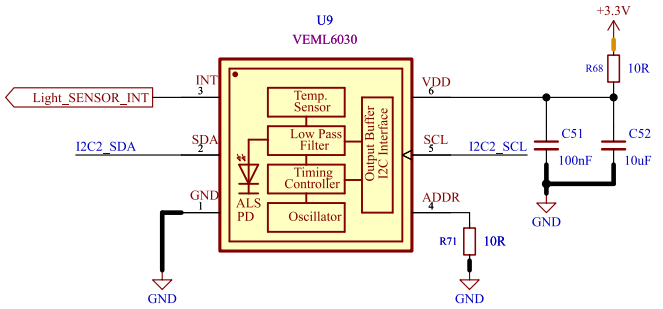
Test Points



BME280 : Temperature, humidity and Pressure sensor



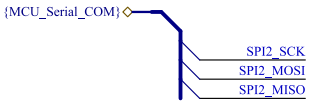
VEML6030 : Light sensor



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Sensor - Environment	File Name: Sensor - Environment.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensor - Environment/		Reviewer:	Size: A4	Sheet: 17 of 22

[18] Sensor - IMU

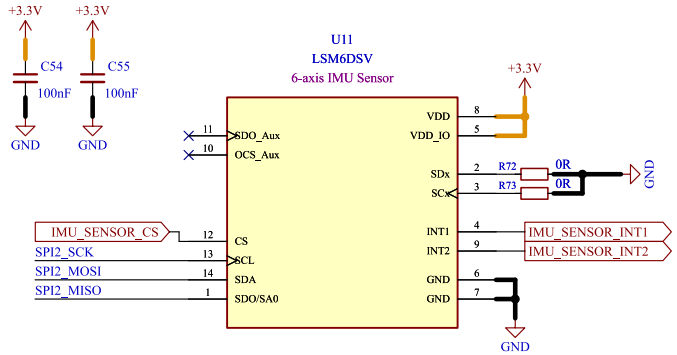
BUS



Test Points



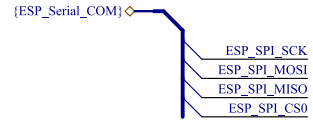
BME280 : IMU Sensor (Accelerometer and Gyroscope)



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Sensor - IMU	File Name: Sensor - IMU.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensor - IMU/		Reviewer:	Size: A4	Sheet: 18 of 22

[19] Memory

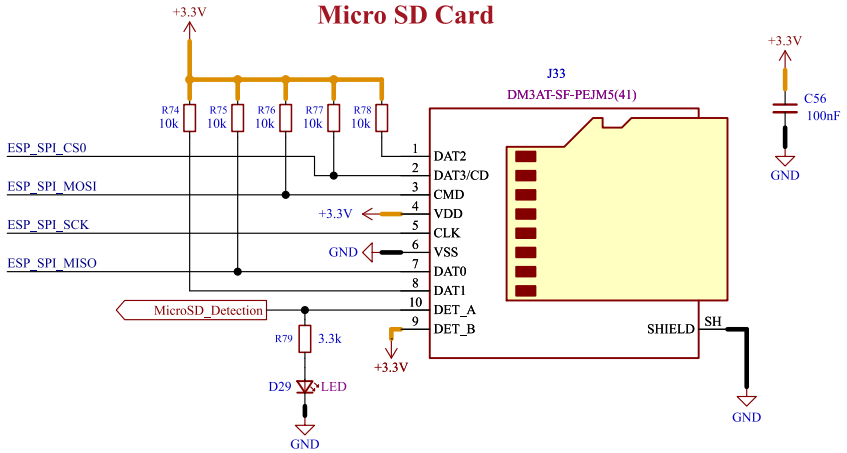
BUS



Test Points



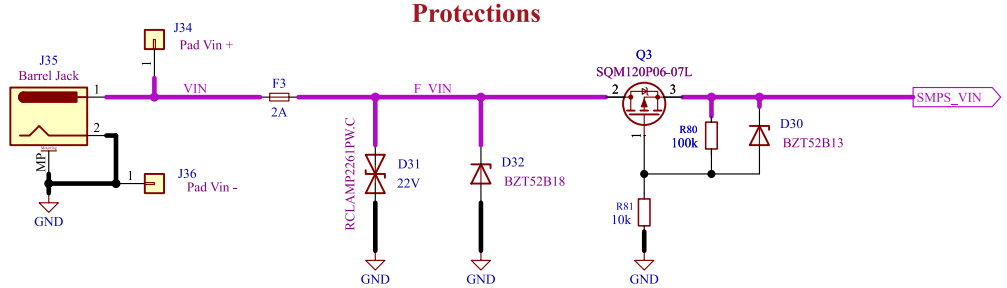
Micro SD Card



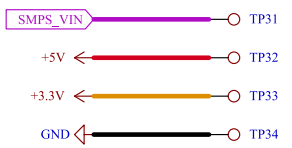
	Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 2499663
		Board Name: DAQ Core	Project Name: Data Acquisition Board	
	Sheet Title: Memory	File Name: Memory.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12 / Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Memory/		Reviewer:	Size: A4 / Sheet: 19 of 22

[20] Power

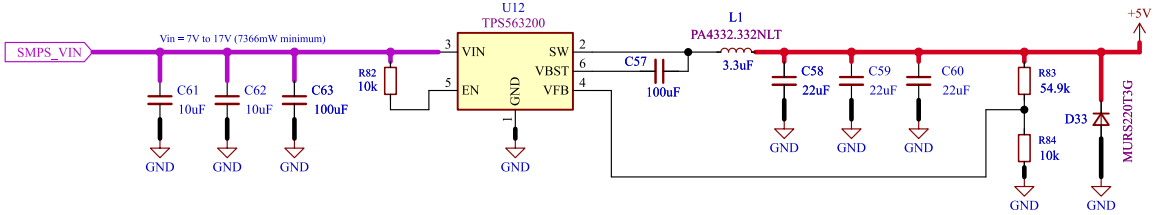
Protections



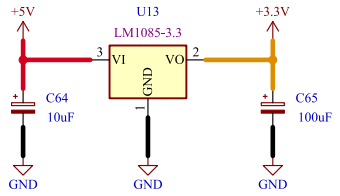
Test Points



5V SMPS Regulator



3V3 LDO Regulator



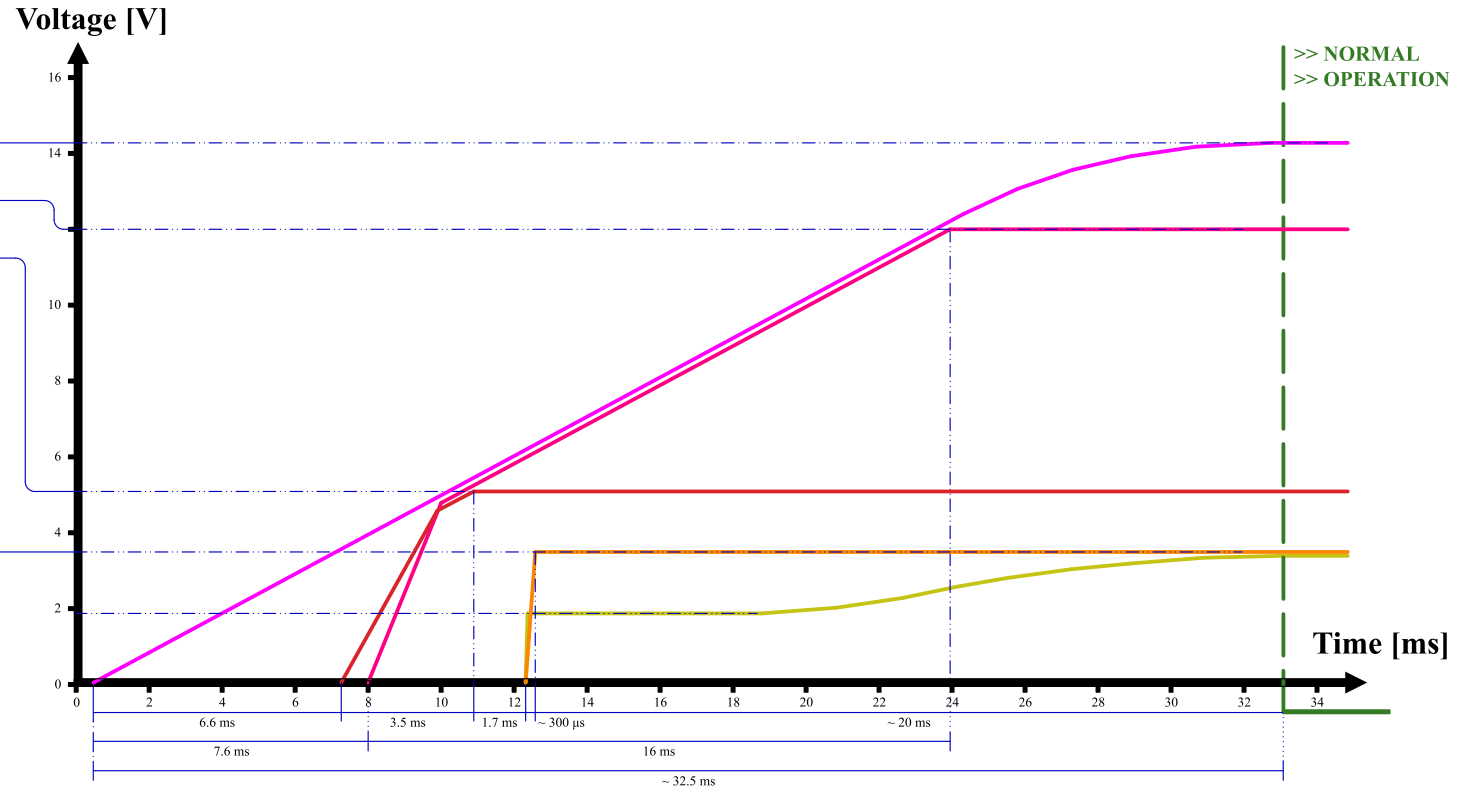
Protections:

->Overvoltage
->Voltage spike
->Overcurrent
->Reverse Polarity

Comments: LM1085-3.3 Datasheet; TPS563200 Datasheet	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Power	File Name: Power.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power/		Reviewer:	Size: **A4**	Sheet: **20** of **22**

[21] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 2499663
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 21 of 22

[22] Revision History

Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 2499663
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **22** of **22**